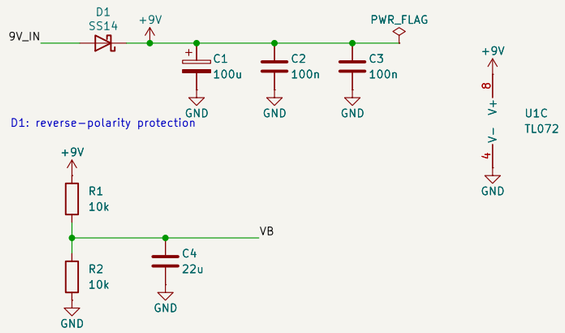
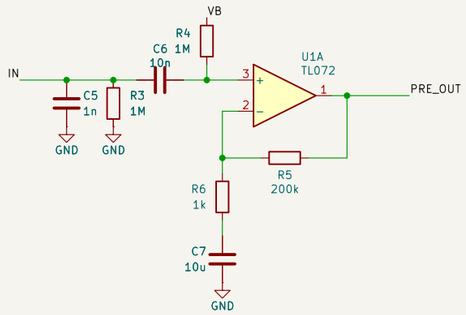
POWER & BIAS



VB = 4.5V virtual ground for U1

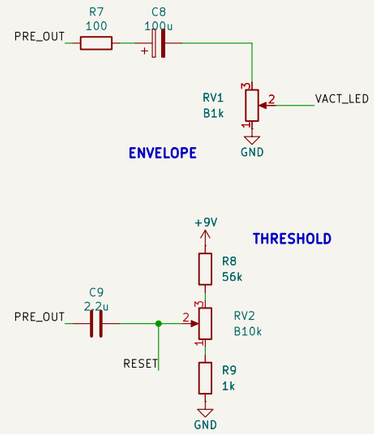
INPUT & PREAMP (U1A)



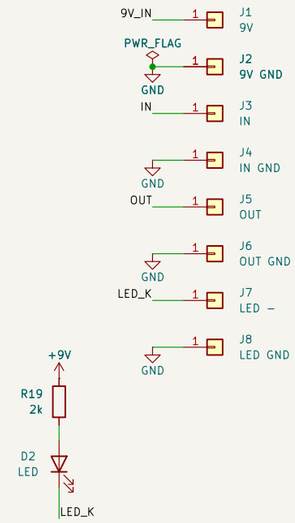
Gain = 1 + R5/R6 = 201 (46 dB)

C7 was 220n (720 Hz bass cut)

ENVELOPE & GATE

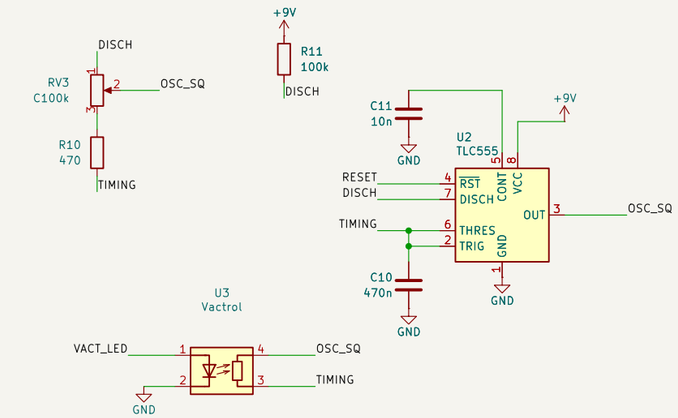


OFF–BOARD WIRING



OSCILLATOR (U2 TLC555)

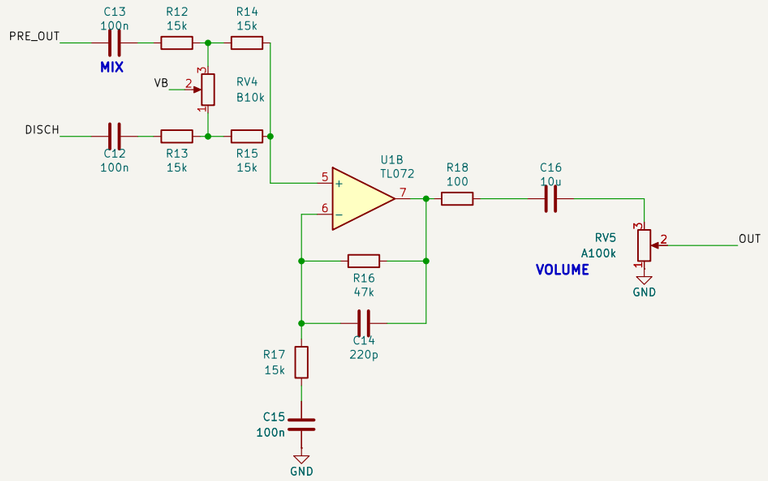
FREQUENCY



LDR in parallel with FREQUENCY:

envelope brightens LED –> pitch rises

MIX & OUTPUT (U1B)



Output gain = 1 + R16/R17 = 4.1

NOTES

1. SMD passives 0805 (hand–solder pads) unless noted; C9 is 1206. LCSC field = JLCPCB part number.
2. Hand–soldered: RV1–RV5 (Alpha 16mm), D2 (3mm LED), U3 (DIY vactrol), J1–J8 wire pads.
3. Changes from Eagle v2: R16 feedback connected, RV3 pin 1 to DISCH, R7/R18/C3/C11 added, C7 220n–>10u, C4 47u elec–>22u MLCC.

Enclosure: Hammond 1590B
Converted from Eagle UglyFace_021 for JLCPCB SMD assembly

Sheet: /
File: UglyFace_03.kicad_sch

Title: Ugly Face 3 – SMD

Size: A3	Date: 2026–09–24	**Rev: 3.0**
KiCad E.D.A. 10.0.4		Id: 1/1